

Instructor Pre-Flight Checklist — AWS Cloud Practitioner (1 Day)

Read this **at least two business days before class**. The course assumes one shared AWS account with N students (typically 8–25), all working in **us-east-2 (Ohio)** through the AWS Management Console. There is no Cloud9 and no local tooling: students need only a modern browser (plus a second browser profile or private window for Lab 1).

1. Account and Student Identities

- **One IAM user per student**, named after the assigned username (`user1` ... `userN`), console access with a password, **no access keys**.
- Attach **AdministratorAccess** (students create IAM users, roles, VPCs, RDS instances, budgets). If your organization forbids that, the minimum is: `IAMFullAccess` , `AmazonEC2FullAccess` , `AmazonVPCFullAccess` , `AmazonS3FullAccess` , `AWSLambda_FullAccess` , `AmazonRDSFullAccess` , `CloudWatchFullAccess` , `AWSBudgetsActionsWithAWSResourceControlAccess` , plus `ce:*` for Cost Explorer.
- Set an **account alias** so the sign-in URL is memorable, and put the URL, username and initial password on a printed card per seat.
- Enable **MFA on the root user**; do not share root.
- Turn **on** the IAM account password policy you want students to see (they only read it).

Student assignment card

Student	Username	Sign-in URL
Alice	user1	https://.signin.aws.amazon.com/console
Bob	user2	...

Username rules: lowercase letters and digits only. Every lab resource is prefixed `userN-` ; S3 bucket names add four random digits for global uniqueness.

2. Service Quotas (request increases 24–48 h ahead)

Check **Service Quotas** → **us-east-2** for each item. Numbers assume N students plus a margin of 2.

Quota	Service	Default	Needed	Why
VPCs per Region	VPC	5	N + 2	Lab 2: one VPC per student
Internet gateways per Region	VPC	5	N + 2	Created with each VPC
Security groups per Region	VPC	2,500	fine	—
Network ACLs per VPC	VPC	200	fine	—
Running On-Demand Standard (A, C, D, H, I, M, R, T, Z) instances (vCPUs)	EC2	often 5 on newer accounts	4 × N	Labs 3–4: 2 × t3.micro (2 vCPU each) per student
AMIs per Region	EC2	50,000	fine	—
Key pairs	EC2	5,000	fine	—
DB instances	RDS	40	N + 2	Lab 7
DB subnet groups	RDS	50	N + 2	Lab 7
Buckets per account	S3	10,000 (general purpose)	fine	1 per student
Concurrent executions	Lambda	1,000	fine	—
Budgets per account	Budgets	20,000	fine	—

Verify the two that bite:

```
aws service-quotas get-service-quota --service-code vpc --quota-code L-F678F1CE --region us-east-2 \
  --query 'Quota.{Name:QuotaName,Value:Value}' # VPCs per Region
aws service-quotas get-service-quota --service-code ec2 --quota-code L-1216C47A --region us-east-2 \
  --query 'Quota.{Name:QuotaName,Value:Value}' # On-Demand Standard vCPUs
```

3. Account-Level Settings That Must Be Right

Setting	Required state	Lab
S3 Block Public Access (account level)	OFF — otherwise students cannot uncheck it at the bucket level and the static website in Lab 5.4 stays 403. Re-enable after class.	5
Cost Explorer	Enabled (Billing → Cost Explorer → <i>Launch</i>). Takes up to 24 h to populate.	8
Cost allocation tag Owner	Activated (Billing → Cost allocation tags). Takes up to 24 h to appear in Cost Explorer.	8
Free Tier / credits	Note whether the account is on the legacy 12-month free tier or the post-July-2025 credit-based plan; adjust the Module 4 pricing slide talk track accordingly.	—
Support plan	Basic is fine. Mention that full Trusted Advisor checks need Business+.	8
Default VPC in us-east-2	Must exist (students never use it, but Lab 3's "VPC" dropdown shows it and the wizard expects it). Do not let anyone delete it.	2–3
EC2 Instance Connect	Works out of the box for Amazon Linux 2023 with port 22 open from <code>0.0.0.0/0</code> (set in Lab 2). Test it before class; if your organization blocks it, add <code>AmazonSSMManagedInstanceCore</code> to the role instructions in Lab 1 and use Session Manager.	3–7
Lambda Function URLs with auth NONE	Allowed by default; some SCPs block them — test one before class.	6
RDS "Free tier" template	Only appears on free-tier-eligible accounts. Otherwise Lab 7 says <i>Dev/Test</i> ; either way the class uses <code>db.t3.micro</code> / <code>db.t4g.micro</code> .	7

4. Cost Estimate

Per student for the day (us-east-2, cleaned up by 4 PM): 2 × t3.micro for ~5 h, 1 × db.t3.micro for ~2 h, EBS volumes and one AMI snapshot, negligible S3/Lambda — ****under 1perstudent***

*.A forgotten RDS instance costs 0.50/day, so **sweep the account after class** (Section 8).

5. Timing Guidance and Where to Cut

The day is scheduled at 225 min labs / 105 min theory. Every theory block is 5–12 minutes; **do not let Module 2 sprawl** — the deck is built so each 2A–2E block is 3–5 content slides.

If you are running late by...	Cut this
10 min	Lab 3 Exercise 3.3 (look around) and Lab 7 Exercise 7.5 (backups tour) — narrate from your screen
20 min	Lab 6 Exercise 6.4 (Function URL) and Lab 8 Exercise 8.3 (rightsized) become instructor demos
30 min	Lab 4 Exercise 4.3 (second server from AMI) becomes a demo

Never cut Lab 8 cleanup. If time is truly gone, tell students to stop and run the sweep yourself (Section 8).

Lab 7 is deliberately split around the afternoon break: students kick off RDS creation (5 min), break for 15, and return to an *Available* database. Do not move the break.

AMI creation (Lab 4.2) takes 2–4 minutes; have students read ahead to 4.3 rather than stare at the console.

Labs are intentionally minimal — core task only, no stretch goals. Resist adding "one more thing" on the day; the 70/30 split only holds if every lab finishes inside its slot.

6. Known Failure Points and Fixes

Symptom	Cause	Fix
Lab 3: instance has no public IP / unreachable	"VPC and more" wizard does not enable auto-assign public IPv4	Lab 2.2 step 1 — verify both subnets
Lab 3: Instance Connect "Failed to connect"	No SSH rule on the SG, no public IP, or wrong subnet	Check Security and Networking tabs
Lab 3: <code>aws s3 ls</code> fails on the instance	IAM instance profile not selected at launch	Actions → Security → Modify IAM role
Lab 4: page will not load	Using <code>https://</code> , or stale public IP after a stop/start	Use <code>http://</code> and the current IP
Lab 4: web-2 shows web-1's instance ID	User data not pasted or typo	Re-launch with the user data
Lab 5.3: website still 403	Account-level Block Public Access is ON	Section 3 — must be off before class
Lab 5.1: "bucket name already exists"	Global namespace collision	Different digits
Lab 6.4: Function URL returns 403	Auth type left at AWS_IAM	Recreate with NONE
Lab 7.2: cannot create database	Subnet group does not span 2 AZs	Both subnets in 7.1
Lab 7.4: <code>mysql</code> times out	DB security group is <code>default</code> instead of <code>userN-db-sg</code>	Modify → Connectivity → fix the SG
Lab 8: "Delete VPC" refuses	RDS still deleting (its network interfaces linger a few minutes)	Wait, retry
Anything: "not authorized"	Signed in as <code>userN-dev</code> (Lab 1 private window)	Check identity top-right

7. Room Setup on the Day

- Deck: open `presentations/aws_cloud_practitioner.html` in Chrome or Edge (it loads Reveal.js from a CDN, so it needs internet). `F` full screen, `O` overview, `S` speaker notes. Slides auto-fit to the screen. Section dividers and lab callout slides mark every hand-off to a lab.
`pdf/aws_cloud_practitioner_slides.pdf` is the offline copy.
- Have your own `user0` identity so you can demo every lab live on the projector.
- Pre-create nothing for students — the point is that they build it.
- Keep the [Lab 8 cleanup table](#) on screen for the last 12 minutes.

8. Post-Class Sweep

Run from CloudShell (admin) after students leave. It lists what is left; delete anything still carrying a `userN` name.

```

R=us-east-2
echo "== EC2 instances (non-terminated) =="; aws ec2 describe-instances --region $R \
  --filters Name=instance-state-name,Values=pending,running,stopped,stopped \
  --query 'Reservations[].Instances[].[Id:InstanceId,Name:Tags[?Key==`Name`]|[0].Value,State:State.Name]' --output table
echo "== Auto Scaling groups =="; aws autoscaling describe-auto-scaling-groups --region $R --query 'AutoScalingGroups[.AutoScalingGroupName'
echo "== Load balancers =="; aws elbv2 describe-load-balancers --region $R --query 'LoadBalancers[.LoadBalancerName'
echo "== AMIs =="; aws ec2 describe-images --owners self --region $R --query 'Images[.Id:ImageId,Name:Name]' --output table
echo "== Snapshots =="; aws ec2 describe-snapshots --owner-ids self --region $R --query 'Snapshots[.Id:SnapshotId,Desc:Description]' --output table
echo "== Volumes (available = unattached) =="; aws ec2 describe-volumes --region $R --filters Name=status,Values=available --query 'Volumes[.VolumeId'
echo "== Elastic IPs =="; aws ec2 describe-addresses --region $R --query 'Addresses[.PublicIp'
echo "== RDS =="; aws rds describe-db-instances --region $R --query 'DBInstances[.DBInstanceIdentifier'
echo "== RDS snapshots =="; aws rds describe-db-snapshots --region $R --snapshot-type manual --query 'DBSnapshots[.DBSnapshotIdentifier'
echo "== VPCs =="; aws ec2 describe-vpcs --region $R --query 'Vpcs[?IsDefault==`false`].[Id:VpcId,Name:Tags[?Key==`Name`]|[0].Value]' --output table
echo "== Lambda =="; aws lambda list-functions --region $R --query 'Functions[.FunctionName'
echo "== DynamoDB =="; aws dynamodb list-tables --region $R --query 'TableNames'
echo "== S3 buckets =="; aws s3 ls | grep -E ' user[0-9]+'
echo "== Budgets =="; aws budgets describe-budgets --account-id $(aws sts get-caller-identity --query Account --output text) --query 'Budgets[.BudgetName'
echo "== IAM leftovers =="; aws iam list-users --query 'Users[?starts_with(UserName,`user`) && contains(UserName,`-dev`)].UserName'; \
  aws iam list-groups --query 'Groups[?contains(GroupName,`-developers`)].GroupName'; \
  aws iam list-roles --query 'Roles[?starts_with(RoleName,`user`)].RoleName'; \
  aws iam list-policies --scope Local --query 'Policies[?starts_with(PolicyName,`user`)].PolicyName'

```

Then: re-enable account-level S3 Block Public Access, and disable or delete the student IAM users until the next run.

9. Content Notes

- The legacy `Slides.pdf` is CLF-C01 material (60 questions, old domain weights, Classic Load Balancer, OpsWorks, per-hour billing, eventual S3 consistency, five pillars) — background reading only; do not present from it. Exam facts and domain mapping are in the README.
- Current facts baked into the deck: six Well-Architected pillars; S3 strong consistency; per-second EC2 billing; ALB/NLB/GWLB; Savings Plans; IAM Identity Center; Sustainability pillar; Snowmobile retired.
- Language: describe the course level as "introductory" in all materials and when speaking to the class.